

CMPE 49T Final Take-home Exam

Q1) - UNet: Medical Segmentation

(a) Encoder Convolution:

Input Matrix:

$$X = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 2 & 2 & 1 & 0 & 0 \\ 1 & 2 & 3 & 2 & 1 & 0 \\ 0 & 1 & 2 & 2 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Encoder Kernel K_{enc} :

$$K_{enc} = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$$

* We perform convolution with stride 2. We calculate the output for each 2×2 block.

$y_{i,j} = \text{sum}(X_{\text{block}} \odot K_{enc})$ where the operation is element wise multiplication followed by summation.

Position (0,0): Block $\begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}$ Result: $(0 \cdot 1) + (1 \cdot -1) + (1 \cdot 1) + (2 \cdot 0) = 0$

Position (0,1): Block $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ Result: $(1 \cdot 1) + (0 \cdot -1) + (2 \cdot 1) + (1 \cdot 0) = 1 + 2 = 3$

Position (0,2): Block $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ Result: 0

2

Ali Ayhan Gunder
2021400219

Position (1,0): Block $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ Result: $(1,1) + (2,-1) + (0,1) + (1,0) = -1$

Position (1,1): Block $\begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix}$ Result: $(3,1) + (2,-1) + (2,1) + (2,0) = 3$

Position (1,2): Block $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$ Result: $(1,1) + (0,-1) + (1,1) + (0,0) = 1+1=2$

Position (2,0): Block $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ Result: 0

Position (2,1): Block $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ Result: $(1,1) + (1,-1) + (0,1) + (0,0) = 1-1=0$

Position (2,2): Block $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ Result: $(1,1) + (0,-1) + (0,1) + (0,0) = 1$

feature map: $F(3 \times 3) =$

$$F = \begin{bmatrix} 0 & 3 & 0 \\ -1 & 3 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

3

(b) Decoder Upsampling

Decoder kernel K_{dec} :

$$K_{dec} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

We perform Transpose Convolution with stride 2. Each value f_{ij} scales the kernel matrix, and the results are placed on the grid at position $(2i, 2j)$.

Non-zero contributions:

* $f_{0,1} = 3$. Adds $\begin{bmatrix} 0 & 3 \\ 3 & 3 \end{bmatrix}$ at output rows 0-1, cols 2-3.

* $f_{1,0} = -1$. Adds $\begin{bmatrix} 0 & -1 \\ -1 & -1 \end{bmatrix}$ at output rows 2-3, cols 0-1.

* $f_{1,1} = 3$. Adds $\begin{bmatrix} 0 & 3 \\ 3 & 3 \end{bmatrix}$ at output rows 2-3, cols 2-3.

* $f_{1,2} = 2$. Adds $\begin{bmatrix} 0 & 2 \\ 2 & 2 \end{bmatrix}$ at output rows 2-3, cols 4-5.

* $f_{2,2} = 1$. Adds $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$ at output rows 4-5, cols 4-5.

Reconstructed Row Output ($\hat{Y}_{row}$): Summing contributions

$$\hat{Y}_{row} = \begin{bmatrix} 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 3 & 0 & 0 \\ 0 & -1 & 0 & 3 & 0 & 2 \\ -1 & -1 & 3 & 3 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

4 Final Prediction $\hat{Y}$ (Threshold > 0):

Ali Ayhan Gunder
2021400219

$$\hat{Y} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

(c) Metrics Calculation:

Ground Truth (Y_{true}): 18 positive interactions.

Predicted ($\hat{Y}$): 13 positive interactions.

Confusion Matrix:

True Positives (TP) = 6 (Coordinates where both are 1)

False Positives (FP) = 6 (Predicted 1, actually 0)

False Negatives (FN) = 12 (Predicted 0, actually 1)

True Negatives (TN) = 12 (Both 0)

$$\text{Total pixels} = 6 + 6 + 12 + 12 = 36$$

Metrics:

* Pixel Accuracy: $\frac{TP+TN}{N} = \frac{6+12}{36} = 0.50$

* IoU (Intersection over Union): $\frac{TP}{TP+FP+FN} = \frac{6}{6+6+12} = 0.25$

* Dice Coefficient: $\frac{2TP}{2TP+FP+FN} = \frac{12}{30} = 0.40$

* Precision: $\frac{TP}{TP+FP} = \frac{6}{12} = 0.50$

* Recall (Sensitivity): $\frac{TP}{TP+FN} = \frac{6}{18} = 0.33$

(d) Dice vs IoU for small Lesions:

- Dice is often preferred over IoU for medical segmentation of small lesions because:

1- Gradient Stability: The gradient of the Dice Loss term is often more stable and less prone to vanishing when the target is very small compared to the background, whereas the IoU loss denominator can fluctuate more aggressively.

2- Metric Sensitivity: While they are monotonically related, Dice naturally weights the "hits" (TP) twice in the numerator and denominator, which can make it slightly more forgiving and interpretable as a harmonic mean of precision and recall, focusing purely on the positive class performance which is critical in class-imbalanced medical tasks.

(e) Boundary Uncertainty Modeling:

- Let us flip one false Positive boundary pixel to be a True Negative (i.e., we correct a mistake)

$$* FP \rightarrow FP - 1$$

$$* IoU \text{ Change: } \frac{TP}{TP + FP + FN} \rightarrow \frac{TP}{TP + (FP - 1) + FN}$$

$$* Dice \text{ Change: } \frac{2TP}{2TP + FP + FN} \rightarrow \frac{2TP}{2TP + (FP - 1) + FN}$$

- Since $2TP > TP$, the denominator in Dice is larger relative to the subtraction of 1. Therefore, the relative percentage change in Dice is smaller than in IoU. This makes Dice slightly less sensitive to single-pixel boundary noise, providing a more robust metric when valid ground truths may vary due to inter-rater variability. (e.g., different radiologists drawing slightly different contours.)

6 (f) Multi-class Extension

- If Y_{true} contains two tumor grades (Class 1 and Class 2)

1- We treat this as two binary segmentation problems.
(one-vs-rest)

2- Class 1 IoU: Compute IoU treating Class 1 as positive and everything else as background.

3- Class 2 IoU: Compute IoU treating Class 2 as positive and everything else as background.

4- Macro-IoU: Take the unweighted average: $\frac{IoU_1 + IoU_2}{2}$.

Q2- RetinaNet for Lesion Detection

(a) Anchor IoU Computation:

Ground Truth Box B_{gt} : (2, 3, 5, 6) (x_1, y_1, x_2, y_2)

Width = $5 - 2 = 3$, Height = $6 - 3 = 3$

Area (B_{gt}) = $3 \times 3 = 9$

- Anchor A_1 (1, 3, 4, 6):

- Width = $4 - 1 = 3$, Height = $6 - 3 = 3$, Area = 9

- Intersection: $x \in [\max(2, 1), \min(5, 4)] = [2, 4]$ ($w=2$)

$y \in [\max(3, 3), \min(6, 6)] = [3, 6]$ ($h=3$)

- Intersection Area = $2 \times 3 = 6$

- Union = $9 + 9 - 6 = 12$

- IoU = $6/12 = 0.5$

7

Anchor A2 (2, 2, 6, 7):

- Width = $6 - 2 = 4$, Height = $7 - 2 = 5$. Area = 20
- Intersection: $x \in [2, 5]$ ($w=3$), $y \in [3, 6]$ ($h=3$)
- Intersection Area = 9 (GT is fully inside)
- Union = $20 + 9 - 9 = 20$
- IoU = $9/20 = 0.45$

Anchor A3 (0, 0, 8, 8):

- Width = 8, Height = 8, Area = 64
- Intersection Area = 9
- Union = $64 + 9 - 9 = 64$
- IoU = $9/64 \approx 0.14$

(b) Focal Loss Computation:

Parameters: $\alpha = 0.25$ (for class 1) $\gamma = 2$ Formula: $FL(p_t) = -\alpha_t (1 - p_t)^\gamma \log(p_t)$

Anchor A1 (Positive, IoU 0.5):

- Label $y = 1$, Prediction $p = 0.7$, So $p_t = 0.7$, $\alpha_t = 0.25$
- Loss = $-0.25 \cdot (1 - 0.7)^2 \cdot \log(0.7)$
 $= -0.25 \cdot (0.3)^2 \cdot (-0.357)$
 $= 0.25 \cdot 0.09 \cdot 0.357 \approx 0.008$

8 Anchor A2 (Negative, IoU 0.45 - treated as Neg)

- Label $y=0$. Prediction $p=0.9$. So $P_t = 1 - 0.9 = 0.1$

$\alpha_t = 0.75$.

$$- \text{Loss} = -0.75 \times (1 - 0.1)^2 \times \log(0.1)$$

$$= -0.75 \cdot (0.81) \cdot (-2.30) \approx 1.4 \text{ (High loss due to high confidence error - "Hard negative")}$$

Anchor A3 (Negative, IoU 0.14)

- Label $y=0$. Prediction $p=0.2$. So $P_t = 1 - 0.2 = 0.8$

$\alpha_t = 0.75$.

$$- \text{Loss} = -0.75 \cdot (1 - 0.8)^2 \cdot \log(0.8)$$

$$= -0.75 \cdot (0.04) \cdot (-0.223)$$

$$= \approx 0.0067 \text{ (Suppressed "Easy Negative")}$$

(C) focal Loss and Lesion Prevalence:

- As lesion prevalence approaches 0%, the dataset becomes dominated by "Easy Negatives" (back-ground). A standard Cross-Entropy loss would sum up thousands of small loss values (like ≈ 0.2) from these easy backgrounds, which collectively overwhelm the gradient of the rare positive lesions. focal Loss introduces the modulating factor $(1 - P_t)^\gamma$. For well-classified backgrounds ($P_t \approx 1$), this factor goes to 0 (e.g. 0.04 in A3), effectively silencing them. This allows the optimizer to focus almost exclusively on the "Hard" examples (like A2) and the actual Positives (A1), solving the extreme class imbalance.

9 (d) Final Predicted Anchor:

- Model Output: The model selects the box with the highest confidence score. Here, A2 has a Lesion Score of 0.9. Thus, the raw model output is A2.

- Justification: Despite A1 having a better IoU and Objectives, the final classification branch "Lesion Score" is the primary sorting key for Non-Maximum Suppression during inference.

- Critique: Ideally, the model is wrong here. It should have scored A1 higher. A2 is a False Positive ($\text{IoU} < 0.5$).

(e) RetinaNet vs YOLO:

- RetinaNet: Designed specifically for dense, imbalanced detection using focal Loss. It is generally more accurate for small objects and small datasets common in medicine, but slower due to the complex feature pyramid and anchor density.

- YOLO (You only Look Once): Optimized for speed and real-time inference. However, early versions struggled with small object clusters. In clinical settings, RetinaNet is preferred for diagnostic radiology (static images, high accuracy needed), while YOLO is standard for interventional videography.

10

(f) Anchor Optimization:

- Standard anchors (e.g. ratios 1:2, 1:1, 2:1) are designed for natural images. Clinical lesions have specific morphologies (e.g. circular nodules, elongated vessels). We can run K-Means Clustering on the width and height of all ground-truth bounding boxes in the training set to find k centroids.

These centroids become the optimal prior anchor shapes, ensuring that the network starts with anchors that statistically match the shapes of the target pathology.

(g) Hard-Negative Mining:

- Easy Negative ($p \approx 0$ for positive class, $P_t \approx 1$): The gradient is scaled by $(1 - P_t)^{\gamma} \approx 0$. The gradient contribution is negligible.

- Hard Negative ($p \approx 1$ for positive class, $P_t \approx 0$): The term $(1 - P_t)^{\gamma} \approx 1$. The gradient is large, similar to standard Cross Entropy.

- Explicitly $\frac{\partial L}{\partial x}$ includes the term $(1 - P_t)^{\gamma}$. This acts as an automated "mining" mechanism, where gradients are only backpropagated for examples the model is currently confused about, ignoring the vast majority of easy background.

Question 3 - Classifier: Confusion Matrix, ROC, Thresholding

(a) Confusion Matrix (Threshold 0.5)

Prediction Table:

Patient	True Label	Prob	Pred (0.5)	Outcome
P1	1	0.90	1	TP
P2	1	0.51	1	TP
P3	1	0.40	0	FN
P4	0	0.20	0	TN
P5	0	0.60	1	FP
P6	0	0.10	0	TN

Confusion Matrix:

	Pred 1	Pred 0
True 1	2 (TP)	1 (FN)
True 0	1 (FP)	2 (TN)

(b) Metrics Computation:

- Sensitivity (Recall): $\frac{TP}{TP + FN} = \frac{2}{2+1} = \frac{2}{3} \approx 0.67$
- Specificity: $\frac{TN}{TN + FP} = \frac{2}{2+1} = \frac{2}{3} \approx 0.67$
- Precision: $\frac{TP}{TP + FP} = \frac{2}{2+1} = \frac{2}{3} \approx 0.67$
- F1 Score: $\frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = 2 \cdot \frac{4/9}{4/3} = \frac{2}{3} \approx 0.67$

12)

(c) ROC-AUC vs IoU in Healthcare

- ROC-AUC: Measures the discriminative ability of a classifier across all possible thresholds. It is ideal for evaluating the general quality of a risk model without committing to a specific clinical operating point.
- IoU (for Classification / Segmentation): Measures the overlap of the positive class at a fixed threshold. It penalizes false Positives and false Negatives equally and heavily. In highly imbalanced tasks (e.g., cancer screening with 1% prevalence), ROC-AUC can sometimes be misleadingly high (dominated by True Negatives), whereas IoU or PR-AUC provides a stricter assessment of the positive class performance.

(d) AUC Computation:

- Compute ROC points at thresholds $\in \{0.7, 0.5, 0.3\}$ and approximate AUC using the trapezoid rule.

Let positives be $\{P_1, P_2, P_3\}$ and negatives be $\{P_4, P_5, P_6\}$.

- Threshold 0.7: Predict positive if $p \geq 0.7$. Then $TP=1$, $FP=0 \Rightarrow (FPR, TPR) = (0/3, 1/3) = (0, 0.33)$
 - Threshold 0.5: $TP=2$, $FP=1 \Rightarrow (FPR, TPR) = (1/3, 2/3) = (0.33, 0.67)$
 - Threshold 0.3: $TP=3$, $FP=1 \Rightarrow (FPR, TPR) = (1/3, 1) = (0.33, 1.00)$
- Including the endpoints $(0,0)$ and $(1,1)$, the AUC is:

$$AUC \approx \frac{1}{2} \cdot \frac{(1/3 + 2/3)}{2} + \frac{2}{3} \cdot \frac{(1+1)}{2} = \frac{1}{6} + \frac{2}{3} = \frac{5}{6} \approx 0.833$$

13

(e) Threshold Sensitivity:

- Threshold 0.4 (High sensitivity Screening):

- P3 (0.40) becomes Positive / TP.

- TP = 3, FN = 0. Sensitivity = 1.0

- P5 is still FP. Specificity = 0.67

- Use case: Rolling out disease (Triage).

- Threshold 0.6 (High Specificity Diagnosis):

- P1 (0.90) is Positive.

- P2 (0.51) and P3 (0.40) become Negative (FN)

- P5 (0.60) becomes Positive? (At exact 0.6 boundary, usually inclusive), If inclusive, FP = 1.

If exclusive, TN = 3. Let's assume inclusive FP = 1

- TP = 1, FN = 2. Sensitivity = 0.33

- Use case: Confirming disease before risky treatment.

(f) Weighted Cross-Entropy:

$$L = -[w_1 y \log(p) + w_0 (1-y) \log(1-p)]$$

Given $w_1 = 3$ (High penalty for missing sick patients) and $w_0 = 1$. Example for Patient P3 (True 1, Pred 0.4)

$$L_{P_3} = -[3 \cdot 1 \cdot \log(0.4) + 1 \cdot 0 \cdot \log(0.6)] = -3 \cdot \log(0.4)$$

$$\approx -3 \cdot (-0.916) = 2.748$$

Standard unweighted loss would be 0.916. The weighting effectively triples the gradient signal for this missed positive, forcing the model to learn this "hard" positive more aggressively.

24 Q4 - CLIP for Radiology Zero-Shot Learning

(a) Similarity Matrix Computation

Image Embeddings (V): $v_1 = [3, 1]$, $v_2 = [0, 4]$.

Text Embeddings (T): $t_1 = [1, 0]$, $t_2 = [0, 1]$.

We compute the dot product similarity matrix $S = V \cdot T^T$.

$$S = \begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^T = \begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix}$$

Detailed scalar products:

$$S_{1,1} = v_1 \cdot t_1 = 3(1) + 1(0) = 3$$

$$S_{1,2} = v_1 \cdot t_2 = 3(0) + 1(1) = 1$$

$$S_{2,1} = v_2 \cdot t_1 = 0(1) + 4(0) = 0$$

$$S_{2,2} = v_2 \cdot t_2 = 0(0) + 4(1) = 4$$

(b) Zero-Shot Softmax Probabilities:

- We apply row-wise softmax to determine the probability of each text matching the given image.

$$p(t_j | v_i) = \frac{\exp(S_{ij})}{\sum_k \exp(S_{i,k})}$$

For Image 1 (v_1):

Logits: $[3, 1]$

$$\exp(3) = 20.086$$

$$\exp(1) = 2.718$$

$$\text{Sum} \approx 22.804$$

$$p(t_1 | v_1) = 20.086 / 22.804 \approx 0.881 \text{ (\% 88.1)}$$

$$p(t_2 | v_1) = 2.718 / 22.804 \approx 0.119 \text{ (\% 11.9)}$$

15 For Image 2 (V_2): Logits $[0, 4]$

$$\exp(0) = 1$$

$$\exp(4) \approx 54.598$$

$$\text{Sum} \approx 55.598$$

$$P(t_1/V_2) = 1/55.598 \approx 0.018 \text{ (1.8\%)}$$

$$P(t_2/V_2) = 54.598/55.598 \approx 0.982 \text{ (98.2\%)}$$

(c) New Image Classification:

New Image: $V_{\text{new}} = [4, 0.5]$

- Similarity to $t_1 [1, 0]$: $4(1) + 0.5(0) = 4$

- Similarity to $t_2 [0, 1]$: $4(0) + 0.5(1) = 0.5$

Since $4 > 0.5$, the model predicts Class 1.

(d) Zero-Shot Learning in Medical AI:

Zero-shot learning allows a model to correctly classify medical conditions it has never been explicitly trained on as a classification target.

Process: The model learns a shared embedding space for images (X-rays) and text (Radiology Reports)

Example: If the model learns that the word "pneumonia" aligns with "cloudy lungs", it can classify a new X-rays as "pneumonia" just by comparing the image vector to the text vector for "pneumonia", even if the training set never had a specific "pneumonia" output neuron.

Benefit: This drastically reduces the need of expensive, expert-annotated class labels for every possible rare disease.

16 (e) Contrastive Loss Gradient:

For a positive pair (v_i, t_i) , the InfoNCE / Contrastive Loss is:

$$L_i = -\log\left(\frac{\exp(v_i \cdot t_i)}{\sum_j \exp(v_i \cdot t_j)}\right)$$

Gradient Behavior:

- The gradient w.r.t the positive text embedding t_i pulls it closer to the image v_i .
- The gradient w.r.t negative text embeddings t_j ($j \neq i$) pushes them away.
- The magnitude is proportional to the hardness; if a negative is very close, the gradient pushing it away is stronger.

(f) Hard Negative Analysis:

Add a semantically similar negative text $t_3 = [0.9, 0.1]$
(very close to $t_1 = [1, 0]$)

New Scores for $v_1 [3, 1]$:

$$1) v_1 \cdot t_1 = 3$$

$$2) v_1 \cdot t_2 = 1$$

$$3) v_1 \cdot t_3 = 3 \cdot (0.9) + 1 \cdot (0.1) = 2.7 + 0.1 = 2.8$$

New Softmax: Logits: $[3, 1, 2.8]$

$$- \exp(3) \approx 20.1$$

$$- \exp(1) \approx 2.7$$

$$- \exp(2.8) \approx 16.4$$

$$- \text{Total Sum} \approx 20.1 + 2.7 + 16.4 = 39.2$$

17

Effect :

$$p(t_1 | v_1) = \frac{20.1}{39.2} \approx 0.51 \text{ (Previously 0.88)}$$

- The presence of a "Hard Negative" steals significant probability mass from the correct class. This highlights the challenge in Medical AI: distinguishing between fine-grained clinical descriptions requires very precise embedding alignment.

Q5 - Vision Transformers in Pathology :

(a) Q, K, V Computations :

Tokens : $p_1 = [1, 0]$, $p_2 = [0, 1]$, $p_3 = [1, 1]$, $p_4 = [2, 1]$

$$W_Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} , W_K = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} , W_V = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

We compute $Q = p W_Q$, $K = p W_K$, $V = p W_V$

Token $p_1 [1, 0]$:

$$- Q_1 = [1, 0] \cdot I = [1, 0]$$

$$- K_1 = [1, 0] \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = [1, 1]$$

$$- V_1 = [1, 0] \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = [1, 0]$$

18

Token $P_2 [0, 1] :$

$$- Q_2 = [0, 1]$$

$$- K_2 = [0, 1] \cdot \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = [0, 1]$$

$$- V_2 = [0, 1] \cdot \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = [1, 1]$$

Token $P_3 [1, 1] :$

$$- Q_3 = [1, 1]$$

$$- K_3 = [1, 1] \cdot \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = [1, 2]$$

$$- V_3 = [1, 1] \cdot \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = [2, 1]$$

Token $P_4 [2, 1] :$

$$- Q_4 = [2, 1]$$

$$- K_4 = [2, 1] \cdot \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = [2, 3]$$

$$- V_4 = [2, 1] \cdot \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = [3, 1]$$

19

(b) Attention Weights for P_4

We computed (scaled) dot-product attention scores for query $Q_4 = [2, 1]$. In a transformer, the standard score is

$$S_{4,j} = \frac{Q_4 \cdot K_j^T}{\sqrt{d_k}}, d_k = 2$$

- Unscaled dot products: $[3, 1, 4, 7]$
- Scaled scores (divide by $\sqrt{2}$):

$$S_4 \approx [2.121, 0.707, 2.828, 4.950]$$

Applying Softmax: $Attn_{4,j} = \frac{e^{S_{4,j}}}{\sum_k e^{S_{4,k}}}$

$$e^{2.121} \approx 8.34$$

$$\text{Sum} \approx 168.6$$

$$e^{0.707} \approx 2.03$$

$$\text{Weights} \approx [0.049, 0.012, 0.100, 0.839]$$

$$e^{2.828} \approx 16.93$$

The token P_4 mostly attends to itself.

$$e^{4.950} \approx 141.3$$

(c) Output Embedding O_4

$$O_4 = \sum_j Attn_{4,j} V_j$$

Using the weights above and $V_1 = [1, 0]$, $V_2 = [1, 1]$, $V_3 = [2, 1]$, $V_4 = [3, 1]$

$$O_4 \approx 0.049 V_1 + 0.012 V_2 + 0.100 V_3 + 0.839 V_4$$

Component-wise:

$$O_4 \approx [0.049 + 0.012 + 0.200 + 2.517, 0 + 0.012 + 0.100 + 0.839] \approx [2.78, 0.95]$$

(d) ViTs and CNN's in Pathology:

Whole Slide Images (WSIs) in pathology are gigapixel images.

* CNNs process local neighborhoods. To connect a tumor region in the top-left to a lymph node in the bottom-right - the receptive field must be excessively deep.

* ViTs use global self-attention. Every patch can "attend" to every other patch in the first layer, allowing the model to instantly capture long-range dependencies, such as the spatial arrangement of stroma vs tumor, which is critical for prognosis.

* Concrete pathology example: In an H&E WSI for colon cancer, one patch may contain tumor glands while a distant patch contains tumor-infiltrating lymphocytes (TILs) at the invasive margin. A ViT can directly link these regions via attention, whereas a CNN would need a very large effective receptive field to relate them.

(e) Sinusoidal Positional Embeddings:

Since the self-attention operation $\text{Attention}(Q, K, V)$ is permutation invariant, we add Positional Embeddings (PE) to inject order.

$$PE_{(pos, 2i)} = \sin\left(\frac{Pos}{10000^{2i/d}}\right), \quad PE_{(pos, 2i+1)} = \cos\left(\frac{Pos}{10000^{2i/d}}\right)$$

21 - for 4 tokens, we generate unique vectors PE_1, PE_2, PE_3, PE_4 based on their integer index positions and add them to the input embeddings before the transformer layers.

- for a minimal numeric example with $d=2$ (so only 1=0 contributes):

$$PE(pos) = [\sin(pos), \cos(pos)]$$

Thus $PE(1) \approx [0.84, 0.54]$ and $PE(2) \approx [0.91, -0.42]$, making token order recoverable by the model.

(f) Multi-Head Attention:

If we add a second head, Head 2 will have a independent weight matrices W_{Q2}, W_{K2}, W_{V2} .

- Head 1 might learn "Shape" (attending to geometric boundaries).
- Head 2 might learn "Texture" (attending to similar cell types).

The outputs $O^{(1)}$ and $O^{(2)}$ are concatenated and projected linearly:
 $O = [O^{(1)} \parallel O^{(2)}] W_O$. This allows the model to capture multiple diverse relationships simultaneously.

(g) Inductive Biases:

- CNNs: Strong inductive bias for locality (pixels nearby are related) and translation invariance (a tumor is a tumor whenever it appears). This makes them data-efficient for small datasets.

- ViTs: Very weak inductive bias. They must learn locality and invariance from massive amount of data. In pathology, where we have millions of patches, this lack of bias is an advantage as it doesn't constrain the model to only look locally.

22 (h) Masked Auto encoder (MAE):

Process: 1. Mask a high percentage of patches.

2. Feed only the visible patches to the Encoder.

3. Feed the encoded latent vectors + learnable "Mask Tokens" to the Decoder.

4. Loss: MSE between reconstructed pixels and original pixels.

- This forces the model to learn a deep understanding of tissue morphology to possible fill in the missing 75.

Concrete pathology example: If an MAE masks out patches covering nuclei-dense tumor epithelium, it must reconstruct the expected H & E texture from surrounding context, which is precisely the kind of presentation that transfers well to downstream tasks like mitosis detection or tumor-vs-stroma segmentation.

Q6 - k-fold Cross-Validation in Medical AI:

(a) Statistical Computation:

Fold Accuracies: $[1.0, 0.5, 1.0, 0.5, 1.0]$

- Mean (μ):

$$\mu = \frac{1.0 + 0.5 + 1.0 + 0.5 + 1.0}{5} = \frac{4.0}{5} = 0.8$$

- Variance (σ^2) and Standard Deviation (σ):

Compute deviations from mean ($x - 0.8$):

$$[0.2, -0.3, 0.2, -0.3, 0.2]$$

Squared deviations:

$$[0.04, 0.09, 0.04, 0.09, 0.04]$$

$$\text{Sum of squares} = 0.04 \times 3 + 0.09 \times 2 = 0.30$$

Population Standard Deviation (for $N=5$):

$$\sigma = \sqrt{\frac{0.30}{5}} = \sqrt{0.06} \approx 0.245$$

(b) Alternating Pattern Interpretation:

The dataset likely contains two distinct subpopulations of patients, for example:

Subpopulation A: "Easy" cases (e.g., clear images, standard disease presentation) Roughly 60% of data.

Subpopulation B: "Hard" cases (e.g., specific scanner artifacts, rare subtypes...). Roughly 40% of data.

— The folds seem to alternate in composition. folds 1, 3, 5 might contain mostly "Easy" test cases, while folds 2, 4 contain "Hard" test cases. This indicates the model is not robust to the domain shift represented by the "Hard" group.

Concrete Imaging example: In chest CT, one group might be standard-dose scans from Scanner Vendor A (easy), while another group might be low-dose or Vendor B with different reconstruction kernels / streak artifacts (hard). Alternating fold performance can happen if folds accidentally concentrate one Vendor / protocol.

24

(C) Patient-Level Splitting:

Mandatory Requirement: In medical imaging, multiple images come from the same patient. These are highly visual correlated.

- **Wrong (Shuffle Split):** If you mix slices from Patient A into both Train and Test, the model memorizes Patient A's anatomy instead of learning the disease features. This leads to Data Leakage and artificially high test accuracy.

- **Right (Group / Patient Split):** All images from patient A must be strictly in either Train or Test. This forces the model to generalize to unseen biological variations.

- **Concrete CT example:** A single chest CT might contain ~ 300 axial slices. If 200 slices land in train and 100 in test, the model can learn patient-specific anatomy and scanner noise texture and appear to generalize while actually memorizing the patient.

(d) Stabilization Methods

To reduce the high variance ($1.0 \leftrightarrow 5.0$)

1) **Stratified Cross-Validation:** ensure every fold has a balance mix of "Easy" and "Hard" cases, or balanced class ratios.

2) **Ensembling:** Train 5 models and average their predictions on test data.

3) **Data Augmentation:** Aggressive color/geometric augmentation to simulate scanner variability and make the model robust to "Hard" cases.

25 (e) Leakage Simulation:

If we performed simple random splitting (patch-level) instead of patient-level splitting:

- Patches from the "Hard" patients would leak into the Training set.
- The model would memorize their texture.
- The fold accuracies for the "Hard" folds (2 and 4) would jump from 0.5 to 0.95.
- Resulting Mean Accuracy would be inflated, masking the model's true clinical failure mode.

Concrete Pathology Example: If patches from the same WSI are split across train/test, the model can exploit slide-specific staining rather than true tumor morphology. This often yields excellent patch-level accuracy but fails on a new hospital's slides.

(f) Bootstrap Confidence Interval:

- We want to estimate the uncertainty of the mean accuracy (0.8)

Method: Resample 5 values with replacement from $[1, 0.5, 1, 0.5, 1]$

Example Resamples:

$$1. [1, 1, 1, 1, 1] \rightarrow \text{Mean } 1.0$$

$$2. [0.5, 0.5, 0.5, 0.5, 0.5] \rightarrow \text{Mean } 0.5$$

...

Standard Error (SE) Estimate:

$$SE = \sigma / \sqrt{N} = 0.245 / \sqrt{5} \approx 0.11$$

$$\mu \pm 1.96 \times SE \approx 0.8 \pm 1.96(0.11) \approx 0.8 \pm 0.216$$

$$CI \approx [0.58, 1.0] \quad \star \text{ This wide interval correctly communicates}$$

with high uncertainty due to small sample size ($N=5$ folds) and high variance.

26

Q7 - Diffusion Models for MRI Denoising

(a) Forward Diffusion Step

- The forward diffusion process progressively adds Gaussian noise to the clean MRI image x_0 according to a variance schedule β_t . The closed-form property allows us to sample x_t at any timestep t directly:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim N(0, I)$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$. As $t \rightarrow T$, $\bar{\alpha}_t \rightarrow 0$, meaning the image x_T becomes pure isotropic Gaussian noise. This simulates the corruption process.

(b) Reverse Denoising Update:

The generative process involves learning to reverse the noise addition. A neural network $\epsilon_\theta(x_t, t)$ is trained to predict the noise component ϵ that was added to reach x_t . The update rule to sample x_{t-1} from x_t is:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right) + \sigma_t z$$

where $z \sim N(0, 1)$ is random noise injected to maintain stochasticity (expected at $t=0$), and σ_t is the variance of the reverse step. By iteratively applying this formula from x_T down to x_0 , the model reconstructs the clean anatomical structure.

27

(C) Linear vs Cosine Noise Schedules

- Linear Schedule: β_t increases linearly from β_{start} to β_{end}

$$\beta_t = \beta_{start} + (\beta_{end} - \beta_{start}) \frac{t-1}{T-1}$$

This can destroy signal quickly early on (large drop in $\bar{\alpha}_t$), risking loss of subtle findings, before the model has learned to preserve them.

- Cosine Schedule: One common choice defines the cumulative signal power directly:

$$\bar{\alpha}_t = \cos^2 \left(\frac{t/T + s}{1+s} \cdot \frac{\pi}{2} \right), \quad s \approx 0.008.$$

Then β_t is derived so that $\bar{\alpha}_t = \prod_{i=1}^t (1 - \beta_i)$, which keeps $\bar{\alpha}_t$ higher in early timesteps and preserves fine anatomy longer.

(d) DDPM vs Classical Denoising:

- Classical Methods: These rely on statistical averaging of similar patches. While effective at removing noise, they often result in smoothing or blurring of edges, which is unacceptable in radiology where edge sharpness defines tumor margins.

- DDPMs (Generative): Instead of just averaging, a diffusion model learns the data distribution of clean MRI scans. When denoising, it effectively "hallucinates" or projects the noisy input back onto the manifold of valid anatomy. This results in plausible, high-frequency texture reconstruction that looks physically realistic, preserving sharpness better than classical filters.

28 Concrete MRI example: In brain MRI denoising, preserving a small multiple-sclerosis lesion boundary or a tiny microbleed matters clinically. Classical denoisers may blur these small hyperintense / hypointense regions; diffusion denoisers can better preserve sharp edges by leveraging learned anatomical priors.

Q8 - Multimodal Fusion: CNN Image + EHR

(a) Fusion Embedding (z)

We have two modality inputs:

V : Image embedding vector (from CNN/ResNet, processing on MRI)

e : EHR embedding vector (from a dense layer processing tabular biomarkers)

- A common early fusion strategy projects them into a shared latent space and sums them.

$$z = W_1 v + W_2 e + b$$

Where W_1 and W_2 transform the specific dimensions of v and e into a common dimensionality d_{shared} . This z now contains a mixed representation of both visual and clinical features.

29 (b) Gating Mechanism (g)

Gating allows the model to dynamically decide which modality is more important for this specific patient.

$$g = \sigma(W_g[v||e] + b_g)$$

- $[v||e]$: Concatenation of image and EHR vectors
- σ : Sigmoid function, keeping outputs in $(0,1)$.
- g : A scalar attention weight.

If $g \approx 1$, the model prioritizes the first component.

If $g \approx 0$, it prioritizes the EHR.

(c) Final risk Prediction:

We use the gate g to weight the contributions before the final classification head:

$$z_{\text{gated}} = g \odot (W_1 v) + (1-g) \odot (W_2 e)$$

$$\hat{y} = \text{Sigmoid}(W_{\text{out}} z_{\text{gated}})$$

Here, the prediction $\hat{y}$ is a weighted combination where the weights g and $(1-g)$ are learned from input context itself.

Concrete numeric Example: Suppose $W_1 = W_2 = I$, $b = 0$, and the model produced $v = [2, 1]$ from chest CT embedding and $e = [0.2, 2]$ from EHR labs. If $W_g = [1, 1, -1, -1]$ and $b_g = 0$ then:

$$g = \sigma(2 + 1 - 0.2 - 2) = \sigma(0.8) \approx 0.69$$

Then:

$$z_{\text{gated}} \approx 0.69v + 0.31e \approx [1.44, 1.31]$$

* This makes it explicit how the gate shifts weight between imaging and EHR for a given patient.

30 (d) Class Imbalance Effects:

- In medical datasets, "sick" patients are often the minority.
- If one modality (e.g. Imaging) is complex and noisy, and the other (EHR) is sparse but clean, an unbalanced model might collapse to ignoring the complex modality entirely, relying only on simple EHR heuristics (e.g. Age 80 implies high risk)
- Gating helps by explicitly modeling the uncertainty; if the image is ambiguous (noisy), the gate can shift weight to the EHR. However, strong regularization (like localized dropout on one modality) is often needed to prevent the model from overfitting to the easier modality.

(e) Interpretability Differences:

- Simple Concatenation: ($z = [v, e]$): This is a "Black Box". The subsequent layers mix the features non-linearly. It is hard to say "The model predicted cancer because of the MRI"
- Gating ($z = gv + (1-g)e$): This is highly interpretable. We can extract the value of g for every patient.
- Patient A ($g = 0.95$): "Model relied 95% on the image"
Radiologist should review the scan.
- Patient B ($g = 0.10$): "Model relied 90% on the lab test"
Clinician should review the bloodwork.
- This transparency builds trust in clinical decision support systems.

31 Q9 - Evaluating Frameworks in Healthcare AI:

(a) Reliability Diagram:

- A reliability diagram checks if the predicted probabilities match the observed event frequencies.

1. Binning: Sort all predictions by probability and divide them into K fixed-width bins (e.g., $[0-0.1], [0.1-0.2]$)

2. Plotting:

- * X-axis (Confidence): Average predicted probability in bin B_k .
- * Y-axis (Accuracy): Fraction of actual positive labels in bin B_k .

3. Perfect Calibration: Points lie on the diagonal $y=x$.

4. Overconfidence: Curve is below the diagonal.

5. Underconfidence: Curve is above the diagonal.

(b) Expected Calibration Error (ECE)

- ECE provides a single scalar summary of the calibration error. It is the weighted average distance between the confidence and accuracy.

$$ECE = \sum_{m=1}^M \frac{|B_m|}{N} |acc(B_m) - conf(B_m)|$$

where N is total samples, $|B_m|$ is number of samples in bin m , $acc(B_m)$ is the true positive rate in the bin, and $conf(B_m)$ is the mean predicted probability. Lower ECE suggests a more trustworthy risk score.

32

Worked mini-example: Suppose 5 patients with predictions $(0.9, 1), (0.8, 0), (0.4, 1), (0.3, 0), (0.2, 0)$, and use 2 bins: $[0, 0.5]$ and $(0.5, 1]$.

High bin $(0.5, 1]$: probs $\{0.9, 0.8\}$, $\text{conf} = 0.85$, $\text{acc} = 0.5 \Rightarrow \text{gap} \approx 0.033$
Contribution: $(2/5) \cdot 0.033 = 0.14$

Low bin $[0, 0.5]$: probs $\{0.4, 0.3, 0.2\}$, $\text{conf} = 0.30$, $\text{acc} = \frac{1}{3} = 0.33$

$\Rightarrow \text{gap} \approx 0.033$ Contribution: $(3/5) \cdot 0.033 \approx 0.02$.

So ECE is $0.14 + 0.02 = 0.16$

(C) Cohen's Kappa (κ):

Accuracy can be misleading when classes are imbalanced. A model predicting "All Healthy" gets 95% accuracy but isn't useful. Kappa measures corrected for chance.

$$\kappa = \frac{P_o - P_e}{1 - P_e}$$

P_o : Observed accuracy

P_e : Expected accuracy by random chance

Values > 0.8 indicate strong agreement. Values ≈ 0 indicate model is no better than random guessing based on class prevalence.

Worked mini-example: Let a test set have $TP=40, FP=10, FN=20, TN=130$. Then $P_o = (TP + TN)/N = 170/200 = 0.85$. Marginals: predicted positive rate $= (TP + FP)/N = 50/200 = 0.25$, predicted negative $= 0.75$
true positive rate $= (TP + FN)/N = 60/200 = 0.30$, true negative $= 0.70$

$$P_e = (0.25)(0.30) + (0.75)(0.70) = 0.075 + 0.525 = 0.60$$

$$\kappa = \frac{0.85 - 0.60}{1 - 0.60} = \frac{0.25}{0.40} = 0.625$$

33

(d) Bootstrap Confidence Intervals

Since we usually have a single test set, a single accuracy number (e.g., 0.85) is just a point estimate. We need to know if 0.85 is statistically significant or just lucky.

- 1- Resample the test predictions N times with replacement.
 - 2- Compute the metric (Accuracy, AUC) for each resampled set.
 - 3- Build a distribution of these N metrics.
 - 4- Report the 2.5th percentile and 97.5th percentile as the 95% Confidence Interval (e.g. $AUC = 0.85 [0.82, 0.88]$)
- This is a standard requirement for FDA approval of medical AI devices.

(e) Clinical Impact of Calibration and Uncertainty:

* Calibration (concrete triage example): For an ED sepsis model, the hospital might set thresholds like $p < 0.10$ = routine care, $0.10 \leq p < 0.30$ = extra vitals, $p \geq 0.30$ = activate sepsis bundle. If the model is overconfident and outputs $p = 0.30$ when the true event rate is 0.10 for similar patients, it triggers unnecessary antibiotics / ICU escalation.

* Uncertainty (concrete imaging example): A chest CT PE detector might output $p = 0.51$ with high uncertainty due to motion artifact or unusual contrast timing. Rather than treating this as a definitive positive, the workflow should route to radiologist review and possibly order confirmatory imaging (repeat CT angiography or an alternative test), reducing harm from borderline false positives.